

4.18 Probability Review (Algebra)

1. Events A and B are independent. It is given that $P(A) = 2P(B)$ and $P(A \cup B) = 0.52$.

Let $x = P(B)$.

- (a) Show that $2x^2 - 3x + 0.52 = 0$. [3]

- (b) Hence find $P(B)$. [2]

- (c) Find $P(A \cap B)$. [1]

2. In a school, 100 students were surveyed about activities A and B .

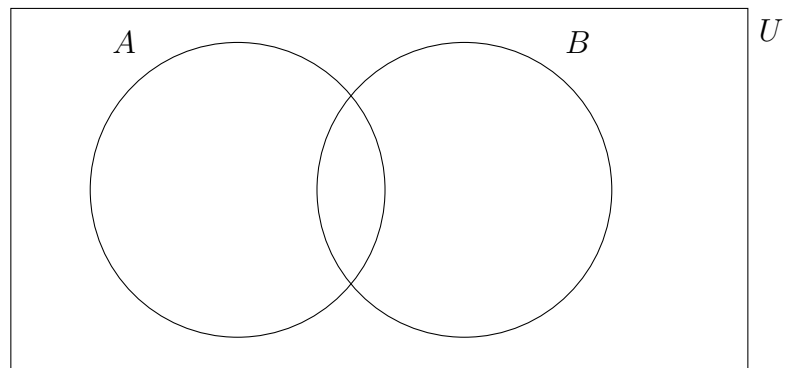
- $P(A) = 0.6$
- $P(B \mid A) = 0.25$
- $P(A' \cap B') = 0.15$

(a) Find $P(A \cap B)$. [2]

(b) Find $P(B)$. [2]

(c) Complete the Venn diagram below, showing the number of students in each region. [2]

(d) Determine whether events A and B are independent. Justify your answer. [2]

[illegible]